Describing Distributions & Linear Regression

ANSWER KEY

Section 1: Interpreting the Least-Squares Regression Line (LSRL)

1 A regression model predicting used car price in dollars from age in years is $\hat{y} = 24000 - 1800x$. Interpret the slope -1800 in context.

a. **For each additional year of car age, the predicted price decreases by 1,800 dollars** ✓

b. A car loses exactly 1,800 dollars every year

c. When the car is 0 years old, its predicted price is $-1,800$ dollars

d. The correlation between age and price is -1800

The slope of an LSRL indicates the predicted (or estimated average) change in the response variable (y) for each 1-unit increase in the explanatory variable (x).

2 In the model $\hat{y} = 45 + 3.2x$, what is the predicted value of y when $x = 10$?

a. 77.0 ✓

b. 48.2

c. 32.0

d. 145.0

Substitute $x = 10$: $\hat{y} = 45 + 3.2(10) = 45 + 32 = 77.0$.

3 Why is the line called the 'least-squares' regression line?

a. **It minimizes the sum of the squared vertical residuals** $\sum(y - \hat{y})^2$ ✓

b. It minimizes the sum of squared distances to the origin

c. It maximizes the square of the correlation coefficient

d. It guarantees all residuals are perfect squares

The LSRL is mathematically derived to minimize the sum of squared vertical differences between observed and predicted values.

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Section 2: Residuals and Residual Plots

4 An observed data point is $(x, y) = (5, 38)$. The regression equation is $\hat{y} = 12 + 6x$. What is the residual for this observation?

a. -4 ✓

b. 4

c. 42

d. -2

Predicted value: $\hat{y} = 12 + 6(5) = 42$. Residual $= y - \hat{y} = 38 - 42 = -4$.

5 What does a positive residual signify?

a. **The actual observed response was greater than the model predicted** ✓

b. The actual observed response was less than the model predicted

c. The regression line has a positive slope

d. The correlation coefficient must be positive

Residual is $\text{actual} - \text{predicted} = y - \hat{y}$. A positive residual means $y > \hat{y}$ (the observed point lies above the regression line).

6 Which pattern in a residual plot indicates that a linear model is appropriate?

a. **A random scatter of points with no discernible pattern around the residual $= 0$ line** ✓

b. A clear curved U-shape pattern

c. A fan shape where residuals expand as x increases

d. Points clustered strictly along an inclined line

A random scatter of residuals around 0 with constant vertical spread indicates that a linear relationship is suitable for the data.

Section 3: Correlation Coefficient r and Coefficient of Determination r^2

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7 If the coefficient of determination is $r^2 = 0.64$ and the slope of the regression line is negative, what is the correlation coefficient r?

a. -0.80 ✓

b. 0.80

c. -0.64

d. -0.4096

$r = \pm\sqrt{r^2} = \pm\sqrt{0.64} = \pm 0.80$. Because the slope is negative, r must share the same sign: $r = -0.80$.

8 What is the correct statistical interpretation of $r^2 = 0.75$?

a. 75% **of the variation in the response variable** y **is explained by the linear relationship with** x ✓

b. 75% of the data points lie directly on the regression line

c. The slope of the regression line is 0.75

d. There is a 75% probability that x causes y

r^2 measures the proportion of variability in the response variable that is accounted for by the linear model with the explanatory variable.

9 Which statement about the correlation coefficient r is FALSE?

a. **If the units of** x **are converted from inches to centimeters,** r **will increase by a factor of** 2.54 ✓

b. r is always between -1 and 1 inclusive

c. Switching the roles of x and y does not change r

d. r measures only the strength and direction of linear relationships

Correlation r is standardized and dimensionless; changing units of measurement has zero effect on r.

Section 4: Outliers, Leverage, and Influential Points

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10 A point that has an extreme x-value relative to the rest of the data is said to have:

a. **High leverage ✓**

b. A large residual

c. High correlation

d. Negative slope

Observations with extreme explanatory (x) values possess high leverage, giving them significant potential to alter the slope of the regression line.

11 What makes a data point an 'influential point' in linear regression?

a. **Removing the point substantially alters the slope, intercept, or correlation ✓**

b. It has a residual equal to 0

c. It lies directly at $(\bar{x}, \bar{y})$

d. It has both positive x and positive y coordinates

An influential point is one whose removal causes a marked change in the regression parameters (slope, intercept, or r).

12 Does a strong correlation between two variables x and y imply that changes in x cause changes in y?

a. **No, correlation does not imply causation due to potential confounding variables ✓**

b. Yes, as long as $|r| > 0.95$

c. Yes, provided the relationship is strictly linear

d. No, unless the sample size is at least 1,000

Association does not imply causation. Observational associations can arise from lurking or confounding variables.

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